Journal of Materials Chemistry A

PAPER

Published on 27 April 2018
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Cite this: J. Mater. Chem. A, 2018, 6, 10348

Microporosity as a new property control factor in graphene-like 2D allotropes†

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We have performed a systematic study on the effect of microporosity on the energy stability and electronic properties of graphene-like materials, including 2D C, B and B–C allotropes. We used a modified crystal structure-search method to yield planar structures by systematically varying microporosity. Our results show that the energy stability of allotropes is strongly correlated with the microporosity and that the trends for C and B allotropes are qualitatively different. The formation energies of the most stable C allotropes with the same microporosity increase quickly with the microporosity, which shows a parabolic relationship. In contrast, B allotropes show a much weaker and linear dependence on microporosity. Our calculations also reveal that basic electronic properties such as metallicity and band gaps also depend strongly on the porosity. The allotropes with low microporosity and low formation energies tend to be metallic, and the semiconducting allotropes appear more often with high microporosity. In contrast to the general trend, our study identified a C allotrope with low microporosity that is quite stable and has a large band gap of 0.58 eV.

Received 21st March 2018
Accepted 27th April 2018

DOI: 10.1039/c8ta02606h

rsc.li/materials-a

1. Introduction

Two-dimensional (2D) materials, such as graphene,[1] borophene,[2] MoS_2,[3] maxene,[4] black phosphorus,[5] 2D organic materials[6] *etc.*, have received great attention for their novel properties over the past few years. They have shown great technological potential for electronic devices,[7,8] solar sails,[9] solar cells,[10] hydrogen separation, gas storage,[11] dehydrogenation[12] and so on. Among all the 2D materials, carbon allotropes have been viewed as prototype materials in both experimental and computational studies. A series of new structures beyond graphene have been proposed, including phagraphene,[13] hexagonal haeckelites,[14] pza-C10,[15] BPC,[11] HOP graphene,[16] pentahexoctite,[17] and net-W.[12] These 2D allotropes consist of carbon rings with various sizes ranging from 3-membered to 12-membered rings. The size of these rings also influences their properties greatly, particularly formation energies and band gaps.[18,19] Similar studies have been conducted for 2D boron allotropes,[20] 2D boron carbides[21] and other compounds.[22,23]

Porosity (the inverse of the 2D atomic density) control is an effective approach to modify the properties and enhance the functions of various materials. Increasing the porosity can also increase the surface area of materials, which may drastically change their electronic properties or their catalytic capability. Microporous materials, such as zeolites and metal–organic frameworks, usually have pore diameters of less than 2 nm. They are particularly interesting since a large portion of their atoms have lower coordination sequence. They have been intensively studied in catalysis and many other areas of chemistry. The importance of microporosity to the properties and stabilities of graphene and related materials has also been noticed.[24–27] It is shown in an earlier study by Z. Wang *et al.* (see ref. 13) that 2D carbon allotropes with lower atomic densities usually have higher formation energies. Compared with 3D materials, the pores in 2D allotropes increase the size of the edges and can also largely change the properties of the allotropes. However, there are no systematic studies of such structural features nor a comprehensive understanding of their effects on the properties of different 2D materials.

In this article, we set out to answer some general questions concerning the effects of microporosity on the stability and properties of mono-atomic layer materials. The questions we

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† Electronic supplementary information (ESI) available: Calculated lattice constants and atomic positions of 1–8 along formation energy curves; the comparison of search results with and without dummy spheres; the phonon spectra of energy edge structures; distribution between energies and microporosities for 8C, 14C, 16C, 12B and 6B6C per unit; all the structure points of all searches from 8 to 16 carbon units and fitting scale values; the distribution between energies and microporosities for the already reported carbon allotropes; the distribution of total sp^2 and sp^2 + sp in 12 carbon atoms per unit; charge difference for 8B8C structures 1 and 4; the results of MD simulation of *Cmmm*-graphene. See DOI: 10.1039/c8ta02606h

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aim to answer are: (1) Is there a general relationship between the microporosity and the formation energy of an allotrope? (2) If so, is the relationship universal or is it different for different materials? (3) What is the effect of the microporosity on the electronic properties of 2D materials, such as the semiconductor band gap or metallicity? We have chosen C, B and B–C allotropes to find answers for these questions. In order to prepare 2D allotropes with varying microporosity, we employ a novel approach that combines the 2D crystal structure search[28,29] and density functional theory (DFT) calculations. The microporosity of the generated structures was controlled by adding dummy spheres that create mesoporous surfaces. Using this method, we have been able to reproduce most of the structures proposed in earlier studies and also generate many new structures by systematically varying microporosity. After analysing the formation energies and properties of these allotropes, we found the important effects of microporosity. Strikingly, we found that C and B allotropes follow qualitatively different correlations between the microporosity and formation energy. Furthermore, we demonstrated that most of the 2D carbon, boron and boron carbide allotropes with relatively low formation energies (relatively low microporosities) are metallic, and the semiconducting allotropes most often have relatively high atomic microporosities.

2. Calculation method

We have performed the structure searches using a particle swarm optimization (PSO) algorithm as implemented in the CALYPSO program.[30,31] In order to control the microporosity, we modified the program. As depicted in Fig. 1a, the initial structures were unbiasedly produced from two-dimensional crystal systems, and all the atomic positions, including dummy spheres, were randomly generated with symmetry and bond

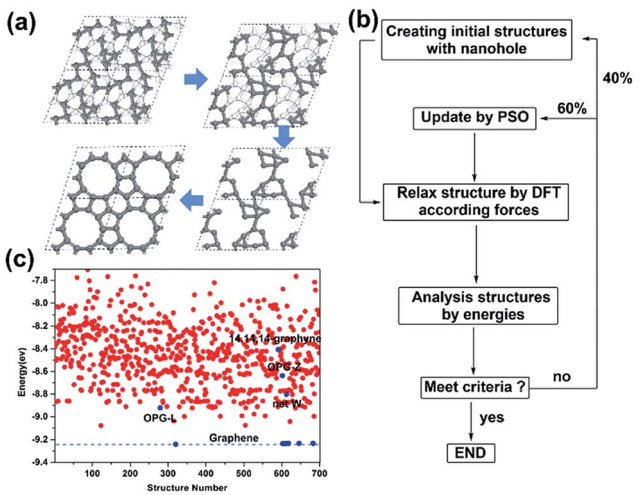


Fig. 1 (a) Process of creating mesopores in a carbon allotrope, white sphere: dummy atom, and black sphere: carbon atom; (b) the process of searching porous allotropes; (c) typical enthalpy evolution of a 12-atom carbon system during an evolutionary structure search and blue spheres show the allotropes that have already been reported.

constraints. In this manner, holes of different sizes were created by controlling the number of dummy atoms. A 15 Å vacuum region was added between parallel slabs to remove interactions. Finally, as shown in Fig. 1b, the initial structures were optimized by DFT calculations and evolved with the PSO and the gene mutation mixed algorithm.[32] All the structures that lost their planar geometry during the optimization, such as becoming multi-layered, were removed. On the other hand, we have kept the structures that, while not perfectly flat, still kept a planar symmetry.

We searched the 2D structures of B, C and B–C with various atoms in each unit cell. The enthalpy evolution of a 12C system is presented in Fig. 1c. All the reported 2D carbon allotropes that match 12 atoms per unit cell were found by our structure search method, including graphene, OPG-L, OPG-Z,[33] net-W,[34] and 14,14,14-graphyne.[35] We also tested our approach in the 2D B, BN and BC_3 systems. All the reported allotropes in previous studies could be reproduced, validating the effectiveness of the method. Furthermore, we also compared our search results with and without using dummy spheres. As shown in Fig. S1,† the use of dummy spheres can successfully generate low energy structures at high microporosity, especially for boron allotropes.

The DFT calculations for structure optimizations were carried out using the Vienna *ab initio* simulation package (VASP).[36–38] The exchange–correlation potential was described by the generalized gradient approximation (GGA) with the Perdew–Burke–Ernzerhof (PBE) functional.[39,40] A plane-wave basis set with an energy cut-off of 600 eV was used to expand the one-electron wave function, and the conjugate gradient (CG) scheme was used to optimize the atomic positions and lattice constants until the force components on each atom were smaller than 0.02 eV $Å^{-1}$. Brillouin zone (BZ) integrations were carried using Monkhorst–Pack[41] sampling grids with a resolution of $2\pi \times 0.1$ $Å^{-1}$. For DOS calculations, Brillouin zone (BZ) sampling grids were improved to $2\pi \times 0.04$ $Å^{-1}$. The hybrid functional methods[42,43] were also used to correct the band structure and the density of states (DOS). The phonon spectrum calculations are obtained by using the PHONOPY package.[44]

The formation energy was defined as the following equation:

$$\Delta E = \frac{E_{\mathrm{tot}}}{n} - E_{\mathrm{ref}}$$

where ΔE, E_{tot}, n and E_{ref} are the formation energy, the total energy of the allotrope, the number of atoms per unit and the energy of the reference structure per atom, respectively. For the convenience of discussion, the microporosity (Φ) was defined as the inverse of the atomic density:

$$\Phi = \frac{1}{\rho} = \frac{A_{\mathrm{tot}}}{n}$$

where Φ, ρ, A_{tot}, and n are the microporosity, atomic density, total area per unit cell and number of atoms per unit, respectively. So the relative microporosity was defined as $\Delta\Phi = \Phi_{\mathrm{tot}} - \Phi_{\mathrm{ref}}$, where Φ_{tot} and Φ_{ref} are the total microporosity and the microporosity of the reference structure, respectively. For each material, we chose the monolayer structure with the lowest energy in our searches as

the reference point of ΔE and $\Delta\Phi$. For example, graphene is taken as the reference allotrope for carbon.

3. Results and discussion

The relative formation energies and the relative atomic microporosities of the 2D allotropes of carbon, boron and boron carbide (1 : 1) which resulted from the PSO structure search are depicted in Fig. 2a–c, respectively. The structures with the lowest energies at different microporosities form an edge in each plot. The phonon spectra of these structures are calculated, and no dynamic instability has been found (Fig. S2†). The representative structures at the energy edges are also shown in the figures. The formation energy edges of all materials increase with increasing microporosity. However, these relationships are strikingly different. For carbon, the formation energy depends on the microporosity nonlinearly (Fig. 2a). It increases faster with microporosity in the low-porosity region (close to graphene), while the trend becomes slower with increasing microporosity. As a matter of fact, the formation energy (ΔE) curve as a function of relative microporosity ($\Delta\Phi$) can be well described with a power function:

$$\Delta E = \alpha^* \Delta\Phi^k \quad (1)$$

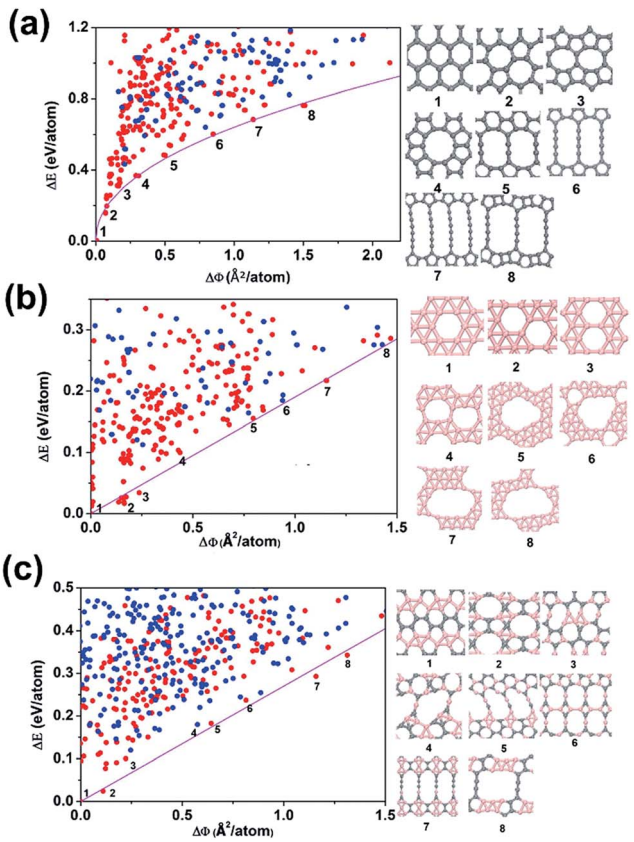


Fig. 2 The relationships between formation energies and relative atomic microporosities for 2D allotropes searched using (a) 12 carbon atoms, (b) 14 boron atoms, (c) 8 boron and 8 carbon atoms. Structures 1–8 are the structures along the lowest formation energy curve. The red and blue dots show the allotropes that are metallic and semiconducting, respectively. The symmetry, the lattice parameters, and the atomic positions can be found in Table S1.†

The best fit of the power k is approximately 0.47, which is very close to 1/2, and the coefficient α has a best fit value of 0.64. Although this relationship is found from the structure search for the unit size of 12C atoms, we found similar relationships while using 8, 10, 14 and 16 carbon atoms per unit (Fig. S3a–d†). As a matter of fact, while we plot all the structure points of all searches from 8 to 16 carbon units, the best-fitted k value becomes 0.5 (Fig. S4†). Interestingly, we found that all the previously reported carbon allotropes are also located on this fitting curve (magenta line in Fig. S5†). Therefore, we conclude that the formation energies of the 2D carbon allotropes along the edge have a square-root dependence on the microporosity.

Different from the 2D carbon allotropes, the edge of the formation energies of the 2D B allotropes depend linearly on the microporosities. Fig. 2b shows the plot of the formation energies and the microporosities of the 2D B allotropes searched using 14B unit cell. Structure 1, with the lowest energy of all the 2D boron allotropes,[20] was also found in our search, and we chose it as the reference structure for all 2D boron allotropes. Fitting the C allotropes to eqn (1), the power k is found to be 1, and the coefficient α is found to be 0.19. It is worth noticing that the energy dependence of B is close to the energy dependence of C on large microporosity. As a matter of fact, if we only consider points from 5 to 8, the C allotrope formation energy edge is also close to a linear dependence on the microporosity. The slope is roughly 0.20 which is very close to that of B. We also depicted the results of searches using 12B atom cells in Fig. S3e.† The formation energy edge also depends linearly on $\Delta\Phi$ with the same slope as that of 14-atom cell results. One structure in Fig. S3e† deviates from the general trend. It has a microporosity of 0.97 and a noticeably low formation energy of 0.13 eV per atom. This structure consists of a honeycomb lattice that is topologically identical to that of graphene. In a recent study, we demonstrated that this structure possesses a Dirac point in its band structure and can be viewed as a true analogy to graphene.[45]

The quick increase in the formation energy for C allotropes at low microporosity can be attributed to the geometry change of the allotropes, which changes the orbital hybridization and weakens the conjugate π bonds. In graphene, the 2D allotrope with the highest atom density, all C atoms are 3-fold coordinated and sp^2 hybridized. The p_z electrons are shared by all carbon atoms and form bonding and anti-bonding π bands. In order to analyse how the bonding features change with increasing microporosity, we marked the allotropes with only sp^2 hybridized C atoms in the energy–porosity plot in Fig. S6.† It shows that most of the low microporosity allotropes consist of only sp^2 hybridized C. Correspondingly, the coordination number remains 3 for all C atoms. In contrast, the allotropes with high microporosity contain both sp^2 and sp hybridized C atoms. The coordination number also changes from 3 to 2 for sp hybridized C atoms. Among the 8 structures located at the edge of the formation energy, structures 1 to 4 consist of only sp^2 hybridized C, whereas structures 5 to 8 consist of both sp^2 and

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sp C atoms. For low-porosity C allotropes (structures 1 to 4 in Fig. 2a), the bond angles of the C atoms at the edges of the holes show large deviations from 120 degrees. Such distortions from the ground state geometry of sp^2 hybrid orbitals increase the energy. In contrast, the high-porosity allotropes contain many sp hybridized C atoms which can accommodate large holes without bond distortion (structures 5 to 8 in Fig. 2a). Therefore, the increase of energy is relatively small. On the other hand, B allotropes are electron deficient. This causes the B atoms to adopt high coordination numbers and form multicenter bonds. As shown in Fig. 2b, the large-porosity structures still maintain high coordination numbers of 4, 5 or 6 for all the B atoms. As a result, the increasing microporosity of B allotropes does not change the formation energy as drastically as that of C allotropes in the region of lower microporosity.

Due to the different behaviours of C and B allotropes with increasing microporosity, it is interesting to see how the formation energy changes with microporosity for 2D boron carbides. We examined this by searching the structures of B–C allotropes with 1 : 1 ratio. The structure searches are performed for cells containing 8C and 8B atoms. The formation energy edge is best fitted by a linear dependence on the microporosity $\Delta\Phi$, which is different from that of C allotropes but similar to that of B allotropes. The slope of the line is found to be 0.27, slightly higher than that of B allotropes. As shown in Fig. 2c, the C atoms in structures 1–3 with low microporosity also maintain sp^2 hybridization and a coordination number of 3. Unlike in C allotropes, the C atoms in B–C do not exhibit large distortions from the sp^2 geometry, due to the existence of B atoms. As a result, the energy does not increase as quickly as that for C allotropes with increasing microporosity. Our calculations suggest that adding B in C allotropes is a good approach to improve microporosity in 2D graphene-like materials.

It is known that many of the structural features of 2D materials such as topology, coordination numbers and symmetry can significantly affect the electronic properties of the materials. However, there are no systematic studies of the relationship between the microporosity and the electronic properties such as metallicity and band gaps for 2D materials. The metallicity of the 2D allotropes is determined by calculating the density of states (DOS) with a fine *k*-mesh. We plot in Fig. 3 the calculated DOS of all the low energy structures for C, B and BC allotropes. For both C and B allotropes, as microporosity increases, the DOS bands become narrower because of the presence of localized states in these structures. However, this does not have a direct effect on the metallicity of the 2D allotropes: almost all the studied structures are metallic, except structures 6 and 8 of the B allotropes and structure 4 of the B–C ones which are semiconductors. In Fig. 2, we have plotted the allotropes in red for metallic and blue for semiconducting. This reveals that a majority of the allotropes with low microporosity and low formation energy are metallic, whereas those with higher microporosity and higher formation energy tend to be semiconducting. As a matter of fact, all the 2D C allotropes with formation energies smaller than 0.42 eV per atom in Fig. 2a are metallic. Similarly, no 2D boron allotrope with a formation energy smaller than 0.11 eV per atom in Fig. 2b has a band gap. In early studies, both experimental and theoretical, almost all the monolayer B allotropes were reported to be metallic.[20,46] These results agree with the fact that semiconducting 2D

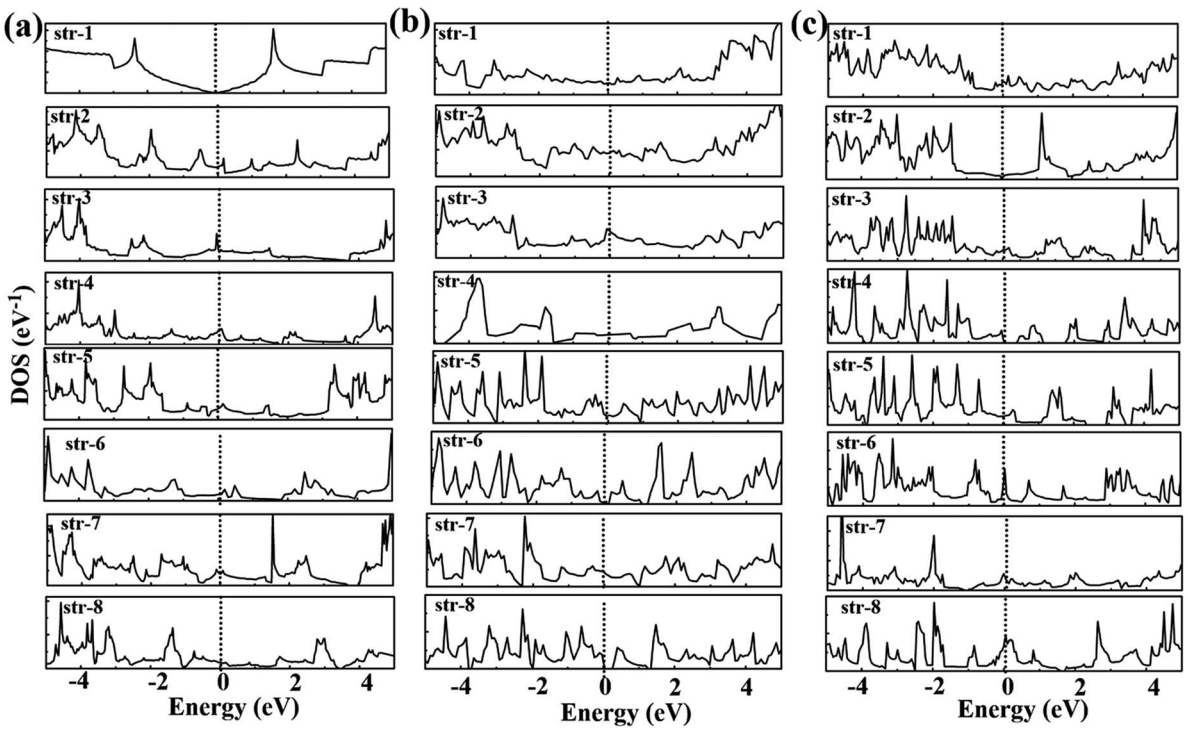


Fig. 3 The density of states of structures 1–8 for (a) 12 carbon atoms per unit, (b) 14 boron atoms per unit, (c) 8 boron atoms and 8 carbon atoms per unit, respectively. The dashed line is the Fermi level.

borophenes are rare. However, our calculations also reveal that the microporosity can increase the chance of forming allotropes with band gaps. Not only more allotropes with high microporosity are semiconducting, but they can also be located at the edge of the formation energy and therefore are among the most stable ones with the same density. Interestingly, compared with 2D carbon and boron allotropes, there seemed to be many more 2D BC allotropes with band gaps. Except for the region where the formation energies are less than 0.11 eV per atom, same as for B, more allotropes present a band gap. This is probably caused by the large charge transfer between the B and C atoms; as an example we calculated the electron transfer for a metal (structure 1) and a semiconductor (structure 4) in 8B8C (see Fig. S7†). The charge difference illustrates that in both cases the electrons transfer from B to C because of their electronegativity difference. QTAIM charge analysis[47] indicates that in structure 1 the charge transfer is 0.62*e* per atom, and in structure 4, it is 1.46*e* per atom.

As clearly shown in Fig. 2a, it is hard to find a 2D carbon allotrope with band gaps, low formation energy and low microporosity. Here, we report a new 2D allotrope (Fig. 4a) which is an exception. Therefore, it is quite promising for future development and application. The structure was found by a PSO search using 14C atoms per unit cell, and therefore it is not included in Fig. 2. It has a space group of *Cmmm* and consists of 5-, 6- and 8-membered rings. There are three types of C–C bonds in the 8-membered rings. The lengths of them are 1.42, 1.50 and 1.51 Å, respectively, the latter two being considerably longer than that of graphene, 1.42 Å. The formation energy of this *Cmmm* structure is 8.98 eV per atom, a value that is only 0.25 eV per atom higher than that of graphene. The stability of this 2D system was confirmed by both the phonon-spectrum (Fig. 4b) and *ab initio* molecular dynamics simulation (Fig. S8†). No imaginary frequencies were found in the phonon spectrum, showing the dynamic stability of the allotrope. The highest optical frequency goes up to 1822 cm^{-1}, which is far higher than those of graphene (~1620 cm^{-1}),[48] blue phosphorene (~540 cm^{-1}) and black phosphorene (~467 cm^{-1}).[49] In order to further examine its thermal stability, *ab initio* molecular dynamics simulations are performed with a 2 × 2 supercell containing 112 atoms using a canonical ensemble (NVT). After being heated at 1500 K for 3 ps with a time step of 1.5 fs, no structural changes occurred, except the expected thermal oscillations of the atoms around their equilibrium positions (Fig. S8b†).

Electronic structure calculations using a hybrid functional (HSE06) show that *Cmmm*-graphene is a direct-gap semiconductor with a band gap of 0.58 eV at the Γ point (Fig. 4c). The band gap obtained using the PBE functional is only 0.18 eV, which is expected since semi-local functionals are known to underestimate semiconductor band gaps. The partial charge calculations of the VBM and CBM states reveal that they are both largely localized at the 8-membered rings (Fig. 4d).

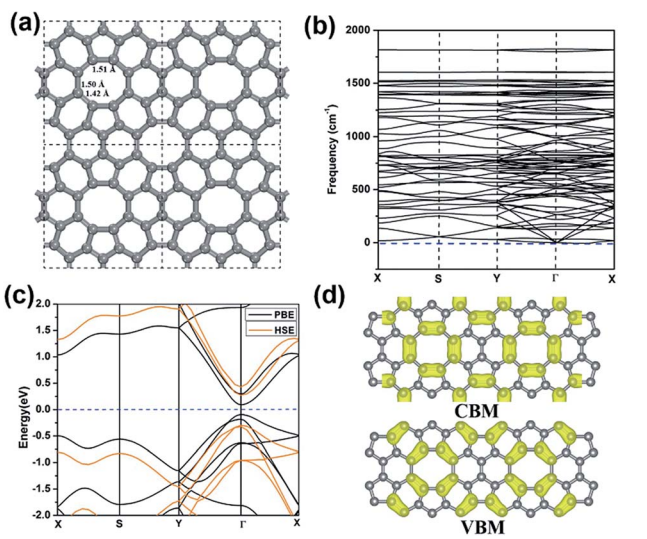


Fig. 4 (a) The structure of *Cmmm*-graphene; (b) phonon dispersion of *Cmmm*-graphene; (c) comparison of band structures from the DFT (black lines) and the HSE (orange lines) model; (d) partial charge density distributions of VBM and CBM of *Cmmm*-graphene.

4. Conclusions

In summary, we have studied the stability and the electronic properties of 2D C, B and BC allotropes with varying mesoporosity. We modified the 2D crystal structure search method based on a PSO algorithm to systematically search the 2D structures with controlled microporosity. Our study revealed striking relationships between the formation energies and the microporosity and found that they are very different for C and B allotropes. The C allotropes exhibit a parabolic relationship, while the B and BC allotropes show a linear dependence. Also, the effect of microporosity on stability is much stronger for the C allotropes than for the B-containing ones. The microporosity also changes the electronic properties: most of the allotropes with low microporosity, both C and B, are metallic, whereas those with high microporosity tend to be more semiconducting. Despite the general trend that low microporosity C allotropes tend to be metallic, we identify an allotrope with a *Cmmm* symmetry which has a large band gap of 0.58 eV and very low microporosity.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

We would like to thank Prof. Ram Seshadri at UCSB for inspiring discussions. This work was supported by the National Natural Science Foundation of China (Grants 21373098) and China Scholarship Council. Part of the calculations are performed on NSF-funded XSEDE resources (TG-DMR130005) especially on the Stampede cluster run by Texas Advanced Computing Center.

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Notes and references

1 K. S. Novoselov, A. K. Geim, S. V. Morozov, D. Jiang, Y. Zhang, S. V. Dubonos, I. V. Grigorieva and A. A. Firsov, *Science*, 2004, **306**, 666–669.

2 A. J. Mannix, X. F. Zhou, B. Kiraly, J. D. Wood, D. Alducin, B. D. Myers, X. Liu, B. L. Fisher, U. Santiago, J. R. Guest, M. J. Yacaman, A. Ponce, A. R. Oganov, M. C. Hersam and N. P. Guisinger, *Science*, 2015, **350**, 1513–1516.

3 W. Li, M. Guo, G. Zhang and Y.-W. Zhang, *Phys. Rev. B*, 2014, **89**, 205402.

4 M. Naguib, V. N. Mochalin, M. W. Barsoum and Y. Gogotsi, *Adv. Mater.*, 2014, **26**, 992–1005.

5 J. Qiao, X. Kong, Z. X. Hu, F. Yang and W. Ji, *Nat. Commun.*, 2014, **5**, 4475.

6 S. Zhao, Y. Wang, J. Dong, C.-T. He, H. Yin, P. An, K. Zhao, X. Zhang, C. Gao, L. Zhang, J. Lv, J. Wang, J. Zhang, A. M. Khattak, N. A. Khan, Z. Wei, J. Zhang, S. Liu, H. Zhao and Z. Tang, *Nat. Energy*, 2016, **1**, 16184.

7 K. I. Bolotin, F. Ghahari, M. D. Shulman, H. L. Stormer and P. Kim, *Nature*, 2009, **462**, 196–199.

8 A. W. Robertson, G. D. Lee, K. He, Y. Fan, C. S. Allen, S. Lee, H. Kim, E. Yoon, H. Zheng, A. I. Kirkland and J. H. Warner, *Nano Lett.*, 2015, **15**, 5950–5955.

9 T. Zhang, H. Chang, Y. Wu, P. Xiao, N. Yi, Y. Lu, Y. Ma, Y. Huang, K. Zhao, X.-Q. Yan, Z.-B. Liu, J.-G. Tian and Y. Chen, *Nat. Photonics*, 2015, **9**, 471–476.

10 K. P. Loh, S. W. Tong and J. Wu, *J. Am. Chem. Soc.*, 2016, **134**, 1095–1102.

11 Q. Song, B. Wang, K. Deng, X. Feng, M. Wagner, J. D. Gale, K. Müllen and L. Zhi, *J. Mater. Chem. C*, 2013, **1**, 38–41.

12 X. Q. Wang, H. D. Li and J. T. Wang, *Phys. Chem. Chem. Phys.*, 2013, **15**, 2024–2030.

13 Z. Wang, X. F. Zhou, X. Zhang, Q. Zhu, H. Dong, M. Zhao and A. R. Oganov, *Nano Lett.*, 2015, **15**, 6182–6186.

14 H. Terrones, M. Terrones, E. Hernández, N. Grobert, J. C. Charlier and P. M. Ajayan, *Phys. Rev. Lett.*, 2000, **84**, 1716–1719.

15 X. Luo, L.-M. Liu, Z. Hu, W.-H. Wang, W.-X. Song, F. Li, S.-J. Zhao, H. Liu, H.-T. Wang and Y. Tian, *J. Phys. Chem. Lett.*, 2012, **3**, 3373–3378.

16 B. Mandal, S. Sarkar, A. Pramanik and P. Sarkar, *Phys. Chem. Chem. Phys.*, 2013, **15**, 21001–21006.

17 B. R. Sharma, A. Manjanath and A. K. Singh, *Sci. Rep.*, 2014, **4**, 7164.

18 W. Liu, Z. F. Wang, Q. W. Shi, J. Yang and F. Liu, *Phys. Rev. B*, 2009, **80**, 233405.

19 J. Bai, X. Zhong, S. Jiang, Y. Huang and X. Duan, *Nat. Nanotechnol.*, 2010, **5**, 190–194.

20 X. Yu, L. Li, X. Xu and C. Tang, *J. Phys. Chem. C*, 2012, **116**, 20075–20079.

21 X. Luo, J. Yang, H. Liu, X. Wu, Y. Wang, Y. Ma, S. H. Wei, X. Gong and H. Xiang, *J. Am. Chem. Soc.*, 2011, **133**, 16285–16290.

22 W. C. Yi, T. Hu, T. Su, R. Islam, M. S. Miao and J. Y. Liu, *J. Mater. Chem. C*, 2017, **5**, 8498–8503.

23 S. Z. Butler, S. M. Hollen, L. Cao, Y. Cui, J. A. Gupta, H. R. Gutierrez, T. F. Heinz, S. S. Hong, J. Huang, A. F. Ismach, E. Johnston-Halperin, M. Kuno, V. V. Plashnitsa, R. D. Robinson, R. S. Ruoff, S. Salahuddin, J. Shan, L. Shi, M. G. Spencer, M. Terrones, W. Windl and J. E. Goldberger, *ACS Nano*, 2013, **7**, 2898–2926.

24 G. Brunetto, P. A. S. Autreto, L. D. Machado, B. I. Santos, R. P. B. dos Santos and D. S. Galvão, *J. Phys. Chem. C*, 2012, **116**, 12810–12813.

25 P. Xu, J. Yang, K. Wang, Z. Zhou and P. Shen, *Chin. Sci. Bull.*, 2012, **57**, 2948–2955.

26 Y. Fang, Y. Lv, R. Che, H. Wu, X. Zhang, D. Gu, G. Zheng and D. Zhao, *J. Am. Chem. Soc.*, 2013, **135**, 1524–1530.

27 Z. Zhuo, X. Wu and J. Yang, *J. Am. Chem. Soc.*, 2016, **138**, 7091–7098.

28 Y. Wang, J. Lv, L. Zhu and Y. Ma, *Phys. Rev. B*, 2010, **82**, 094116.

29 Y. Wang, M. Miao, J. Lv, L. Zhu, K. Yin, H. Liu and Y. Ma, *J. Chem. Phys.*, 2012, **137**, 224108.

30 Y. Wang, J. Lv, L. Zhu and Y. Ma, *Comput. Phys. Commun.*, 2012, **183**, 2063–2070.

31 H. Wang, Y. Wang, J. Lv, Q. Li, L. Zhang and Y. Ma, *Comput. Mater. Sci.*, 2016, **112**, 406–415.

32 S. M. Woodley, P. D. Battle, J. D. Gale and C. R. A. Catlow, *Phys. Chem. Chem. Phys.*, 2004, **6**, 1815.

33 C. Su, H. Jiang and J. Feng, *Phys. Rev. B*, 2013, **87**, 075453.

34 L. C. Xu, R. Z. Wang, M. S. Miao, X. L. Wei, Y. P. Chen, H. Yan, W. M. Lau, L. M. Liu and Y. M. Ma, *Nanoscale*, 2014, **6**, 1113–1118.

35 A. N. Enyashin and A. L. Ivanovskii, *Phys. Status Solidi B*, 2011, **248**, 1879–1883.

36 G. Kresse and J. Hafner, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 1994, **49**, 14251–14269.

37 G. Kresse and J. Furthmüller, *Comput. Mater. Sci.*, 1996, **6**, 15–50.

38 G. Kresse and J. Furthmüller, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 1996, **54**, 11169–11186.

39 P. E. Blöchl, *Phys. Rev. B: Condens. Matter Mater. Phys.*, 1994, **50**, 17953–17979.

40 J. P. Perdew, K. Burke and M. Ernzerhof, *Phys. Rev. Lett.*, 1996, **77**, 3865–3868.

41 H. J. Monkhorst and J. D. Pack, *Phys. Rev. B: Solid State*, 1976, **13**, 5188–5192.

42 J. Heyd, G. E. Scuseria and M. Ernzerhof, *J. Chem. Phys.*, 2003, **118**, 8207.

43 J. Heyd and G. E. Scuseria, *J. Chem. Phys.*, 2004, **121**, 1187–1192.

44 A. Togo, L. Chaput and I. Tanaka, *Phys. Rev. B*, 2015, **91**, 094306.

45 W. C. Yi, W. Liu, J. Botana, L. Zhao, Z. Liu, J. Y. Liu and M. S. Miao, *J. Phys. Chem. Lett.*, 2017, **8**, 2647–2653.

46 B. Feng, J. Zhang, Q. Zhong, W. Li, S. Li, H. Li, P. Cheng, S. Meng, L. Chen and K. Wu, *Nat. Chem.*, 2016, **8**, 563–568.

47 E. Sanville, S. D. Kenny, R. Smith and G. Henkelman, *J. Comput. Chem.*, 2007, **28**, 899–908.

48 S. Piscanec, M. Lazzeri, F. Mauri, A. C. Ferrari and J. Robertson, *Phys. Rev. Lett.*, 2004, **93**, 185503.

49 A. Jain and A. J. McGaughey, *Sci. Rep.*, 2015, **5**, 8501.